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Research project
Solving Unpaired Translation Problems via Diffusion Distillation

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Annotation

Image-to-image translation is a fundamental problem in computer vision, aiming to map images from one domain to another while preserving their underlying content. Modern methods for solving this task are mainly divided into three families with their significant drawbacks: approaches based on diffusion models suffer from slow inference, based on GANs are fast, but have inferior generation quality, and optimal transport based are theoretically justified but are hard to scale. In this work, we suggest modifying the state-of-the-art GAN-based method CUT by replacing the adversarial training with the Distribution Matching technique of diffusion distillation. We qualitatively show that this would synergize the advantages of both approaches, potentially achieving a balance between fast inference and high-quality image generation.

Аннотация

Задача перевода изображений между доменами является одной из ключевых проблем в компьютерном зрении. Её цель — преобразовывать изображения из одного домена в другой, сохраняя их смысловое содержание. Современные методы решения этой задачи делятся на три основные группы, каждая из которых имеет существенные недостатки: диффузионные модели обеспечивают высокое качество, но работают медленно; GAN-методы выполняют преобразование быстро, но уступают в качестве генерации; методы оптимального транспорта имеют теоретическое обоснование, но плохо масштабируются. В данной работе мы предлагаем модифицировать передовой GAN-метод CUT, заменив состязательное обучение на метод Distribution Matching, применяемый для дистилляции диффузионых моделей. Более того, мы демонстрируем, что такой подход объединит преимущества обеих методик, потенциально обеспечив баланс между скоростью вывода и высоким качеством генерации изображений.

Keywords

Unpaired image-to-image translation, distribution matching, diffusion models, contrastive learning

1 Introduction

In this work, we propose an approach to unpaired domain translation, specifically, the unpaired image-to-image problem. This is an important task because its solution allows for domain translation without requiring paired datasets, which are often difficult or expensive to obtain. This flexibility enables practical applications like medical imaging, style transfer, and photo enhancement, where aligned data is scarce or unavailable.

The key challenge is that for training there are no paired samples $(X_0, Y_0) \dots (X_n, Y_n)$, where X_i is an image from the source domain and Y_i is the corresponding image from the target domain. On the contrary, there are two independent datasets $(X_0, \dots, X_n)$ and $(Y_0, \dots, Y_n)$, and the exact mapping between them is unknown. The existing models for solving the problem are divided into three main families, each with its inherent benefits and limitations. Optimal transport-based methods are well theorized but computationally expensive and poorly scaled. GAN-based approaches offer fast inference but often fall short of covering all modes of the distribution. Meanwhile, diffusion-based techniques are relatively slow but deliver superior generation quality. Therefore, we suggest taking a GAN method as a basis and improving it by applying diffusion models while maintaining the speed of generation.

One way to do this is to combine two state-of-the-art approaches in each family of models: Contrastive Unpaired Translation [11] as the basic GAN method and Distribution Matching Distillation [15] as the improving diffusion technique. DMD allows for using the pretrained diffusion as a training signal, which gives realistic pictures. Meanwhile, CUT combines adversarial training with preserving the proximity between pictures by contrastive loss. This substitution aims to compensate for CUT's inferior generation quality while retaining its main advantage being fast generation.

The primary goal of this work is to develop and evaluate the proposed method. The key tasks are as follows. Firstly, rewriting training and evaluation scripts in the codebase to adapt the framework for our specific tasks. Secondly, implementing custom loss functions and making novel architectural modifications. Thirdly, conducting comparative experiments on unpaired datasets to validate performance, where we aim to demonstrate quantitative and qualitative advantages. Fourthly, establishing directions for future optimization.

The paper is organized as follows: Section 1 provides an introduction to the research field and outlines the problems and tasks. Related works are discussed in Section 2. Section 3 suggests the methodology and describes the proposed approach in detail. Section 4 is devoted to a discussion of the current results, followed by a conclusion and future work in Section 5.

2 Literature review

2.1 GAN-based models

For a long time, GANs were a prevalent paradigm for image-to-image translation. They learn a generator that generates images on the inference time and a discriminator that learns to distinguish the real images from the fake ones using adversarial loss. The GAN-based approaches for the considered task mostly can be categorized into two families: one-sided, which require only the target generator, and two-sided, which require both the source and the target networks.

The most popular one-sided models are CycleGAN [18] and DiscoGAN [7]. They are trained using cycle-consistency loss, enforcing that translating an image from the source to the target and back to the source should result in the original image (and vice versa). This ensures meaningful translations that retain the features of the original.

In contrast, CUT [11] is a one-sided method, so it needs only the target generator. It uses the adversarial loss as any other GAN method, but also exploits its own PatchNCELoss for preserving proximity between the input and the output. This loss is responsible for input-output similarity and is calculated using supervised contrastive loss for patches of an image. This method is state-of-the-art among those based on GAN architectures.

However, GAN-based models have inferior performance than diffusion-based approaches, as it was shown in EGSDE [17]. Moreover, as any other GAN architecture, they suffer from mode collapse [1]: the network usually learns only a few modes of the target distribution and ignores the others.

2.2 Diffusion-based models

Diffusion models consider the process of gradual picture noising and generate pictures from noise by reversing this process in time. Approaches based on them are also divided into two types: one-sided and two-sided.

One-sided methods require only the target domain diffusion model. Among them, SDEdit [10] partially stochastically noises the input and denoises it with the target domain network. ILVR [2] guides the translation process by conditioning on the input/reference image. EGSDE extends this idea by introducing the energy function that allows to preserve the domain-independent features and discard domain-specific ones.

Two-sided methods suggest concatenating the source and the target diffusion models. One of the successful instances of this idea is DDIB [14]. It deterministically encodes the input image

into noise with the source model and transforms it into the output with the target model, resulting in a cycle-consistent high-quality mapping.

Although these methods in general have superior generation quality than the GAN-based ones, they suffer from the need to invoke the neural network dozens of times during the generation time, resulting in relatively slow performance.

2.3 Optimal transport based methods

The last family of unpaired image-to-image translation approaches is methods based on optimal transport. OT is a mathematical problem for finding the most efficient way to move mass from one distribution to another. Therefore, optimal transport is a logical formalization of translation between domains problem, so techniques from this group are the most theoretically explained.

One of the popular approaches in this field is Neural Optimal Transport [8]. It exploits the optimal transport idea of finding a mapping with the least transport cost and adds a regularizer that is responsible for making the network outputs coincide with the target distribution. Although it is a theoretically well-explained and scalable method, it suffers from problems similar to GAN, because it is also based on adversarial training. Another area of work in this direction is Schrodinger bridges that try to combine diffusion-based models and optimal transport, but they are computationally expensive, have low scalability [3].

3 Background

As we have discovered, adversarial-based methods for unpaired image-to-image translation are the fastest and computationally cheap, but they suffer from worse image quality and mode collapse. Therefore, this work suggests replacing adversarial loss with more modern methods and giving superior generation quality diffusion distillation methods. The basis of our method is the state-of-the-art GAN-based method CUT [11] and one of the best distillation technique DMD [15]. The first introduces an effective way of measuring the distance between the samples preserving semantic quality and not just the per-pixel similarity. The second uses a pre-trained diffusion model as a discriminator that measures the likelihood of the samples and pushes the generator towards producing realistic samples.

Contrastive Unpaired Translation. CUT uses a contrastive learning framework to ensure that the translated image preserves the content of the source image while adopting the

style of the target domain. It works as is shown on the figure 3.1: for an input image x, the generator G produces a translated image $\hat{y}$. Then x and $\hat{y}$ both are divided into S patches. For each patch, the method extracts features from L selected layers in G's encoder, processing them through a two-layer MLP network H_l to generate feature stacks $z_l = H_l(G^l_{enc}(x))$ for the source image and $\hat{z}_l = H_l(G^l_{enc}(\hat{y}))$ for the output, where G^l_{enc} represents the output of the l-th chosen layer. For each patch for each $\hat{z}^s_l$ there is a corresponding encoded patch in the original image that is denoted as z^s_l and the rest of patches are denoted as $z^{S\setminus s}_l$.

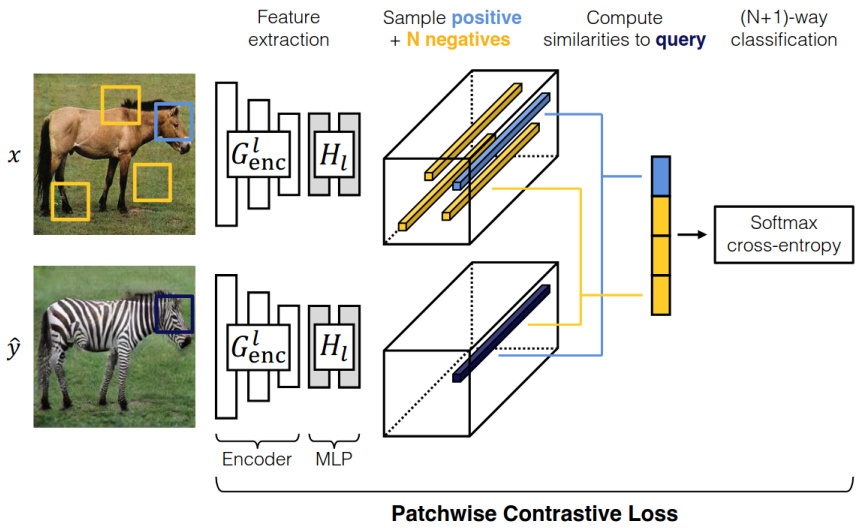

Figure 3.1: A schematic diagram of the PatchNCELoss as presented in the original CUT paper.

Then the generator and discriminator are trained using the following loss:

$$\mathcal{L}_{GAN}(G, D, X, Y) + \lambda_X \mathcal{L}_{PatchNCE}(G, H, X) + \lambda_Y \mathcal{L}_{PatchNCE}(G, H, Y)$$

where $\mathcal{L}_{GAN}(G, D, X, Y)$ is the usual adversarial loss that pushes matching between the output images and the target distribution:

$$\mathcal{L}_{GAN}(G, D, X, Y) = \mathbb{E}_{y\sim Y} \log D(y) + \mathbb{E}_{x\sim X} \log(1 - D(G(x)))$$

and $\mathcal{L}_{PatchNCE}(G, H, X)$ is a measure of input-output similarity calculated as follows:

$$\mathcal{L}_{PatchNCE}(G, H, X) = \mathbb{E}_{x\sim X} \sum_{i=1}^{L} \sum_{i=1}^{S_l} l(\hat{z}^s_l, z^s_l, z^{S\setminus s}_l)$$

where l is supervised contrastive loss:

$$l(v, x, y) = -\log\left[\frac{\exp(v \cdot x/\tau)}{\exp(v \cdot x/\tau) + \sum_{n=1}^{N} \exp(v \cdot y_n/\tau)}\right]$$

To sum up, CUT exploits two ideas: adversarial learning for consistency with the target distribution and PatchNCELoss to measure the similarity of source and target images by comparing neural network representations of patches of these images.

Diffusion models. Diffusion models [5], [13] learn to generate data by consequently denoising images corrupted by a diffusion forward process. It transforms an initial distribution p^0 towards some simple noise distribution with a stochastic differential equation (SDE):

$$dx_t = F(x_t, t)dt + g(t)dw_t$$

where $F(x_t, t)$ is a drift function which represents the velocity vector, $g(t)$ is a diffusion coefficient controlling the magnitude of stochastic forces, and w_t stands for a standard Wiener process.

The same dynamics can be described as an ODE by removing stochasticity:

$$dx_t = \left[F(x_t, t) - \frac{1}{2}g(t)^2 \nabla \log p_t(x_t)\right] dt$$

where $\nabla \log p_t(x_t)$ is a score-function of the marginal distribution p_t at x_t.

The forward ODE and both reverse-time ODE and SDE (for sampling images from a diffusion model) require knowledge of the score function $\log p_t(x_t)$. However, p_t is intractable in practice. Instead, diffusion models learn a score network $s_\phi(x_t, t)$ to approximate the score-function by minimizing the weighted Denoising Score Matching objective:

$$\int_{t=0}^{T} w(t)\mathbb{E}_{\substack{x_0 \sim p^0 \\ x_t|x_0 \sim p_t(x_t|x_0)}} \|s_\phi(x_t, t) - \nabla_{x_t} \log p_t(x_t|x_0)\|_2^2 dt.$$

where $w(t)$ controls the importance of different noise levels and $p_t(x_t|x_0)$ is a conditional transition distribution of the forward diffusion process.

The synthesized samples can then be drawn by reversing the forward process ODE or SDE using the score network $s_\phi(x_t, t)$.

Distribution Matching Distillation. Distribution Matching Distillation distills a teacher diffusion model s^{real} into a one-step generator G_θ. Theoretically, it can be done by minimizing the KL divergence between distributions of the generator and the teacher. However, a diffusion model does not learn the exact score function of the original distribution, but rather an approximation

noised to an arbitrary time level. Moreover, the exact score function can exhibit pathological behavior — if it is defined on a manifold, it vanishes almost everywhere except near the manifold, where its norm blows up to extremely large values. To address these problems, the authors exploit Integral KL [9]:

$$\mathcal{L}_{IKL}(q,p) = \int_{t=0}^{T} w(t)\mathcal{D}_{KL}(q^t, p^t)\, dt = \int_{t=0}^{T} w(t)\mathbb{E}_{x_t\sim q^t}\left[\log \frac{q^t(x_t)}{p^t(x_t)}\right] dt$$

where q^t and p^t denote the marginal densities of the forward SDE initialized with the distributions of the generator and the teacher respectively.

It integrates the KL divergence along a diffusion process, which allows using knowledge from the pretrained diffusion model s^{real} at multiple diffusion time steps.

The gradient of this IKL divergence with respect to the generator parameters can be expressed as the integrated difference between the score functions of the teacher and the generator output distributions:

$$\nabla_\theta \mathcal{L}_{\text{IKL}} = \int_{t=0}^{T} w(t)\mathbb{E}_{\substack{z\sim\mathcal{N}(0,I)\\ x_0=G_\theta(z)\\ x_t|x_0\sim p_t(x_t|x_0)}} \left[s_q^t(x) - s_p^t(x)\right] \frac{\partial x_t}{\partial \theta} dt$$

where s_q^t and s_p^t are score functions of the marginal distributions with densities p^t and q^t. As s_p^t is provided by the pretrained diffusion model s^{real}, the primary task reduces to training an additional diffusion model s_ϕ^{fake} to approximate q^t.

As a result, the training process consists of two alternating phases:

1 Optimizing the generator G_θ by minimizing the IKL divergence loss,

2 Updating the auxiliary diffusion model s_ϕ^{fake} to match the generator's synthetic distribution.

The loss is used for the training is an improved via adaptive weighting and simplified parametrization version of DSM objective being EDM loss [6]:

$$\int_{t=0}^{T} w(t)\mathbb{E}_{\substack{x_0\sim p^0\\ x_t|x_0\sim p_t(x_t|x_0)}} \|D_\phi^{\text{fake}}(x_t,t) - x_0\|_2^2 dt.$$

where $w(t)$ is an adaptive weight based on signal-to-noise ratio (SNR) and $D_\phi^{\text{fake}}(x_t,t)$ is a denoiser parametrization of the diffusion model:

$$s_\phi^{\text{fake}}(x,t) = \frac{D_\phi^{\text{fake}}(x,t) - x}{g^2(t)}$$

4 Methodology

In this section, we introduce our approach. As mentioned before, it suggests replacing the adversarial learning part of CUT loss with DMD loss.

More precisely, we aim to train a generator G that translates images from the source domain to the target. As in the DMD setup, we have a diffusion model pretrained on the target domain s^{real} and an auxiliary trainable diffusion model s^{fake} to approximate the distribution of the outputs of G. Then the generator is updated with the merged loss:

$$\mathcal{L}_{\text{IKL}} + \lambda_{\text{OT}} \mathcal{L}_{\text{PatchNCE}}$$

where λ_{OT} controls the similarity of generator inputs and outputs, and $\mathcal{L}_{\text{PatchNCE}}$ is the CUT loss without the adversarial part. The s^{fake} is trained on EDM loss.

This is a meaningful replacement because DMD is a high-quality framework that ensures hitting the target distribution in general, moreover, it has been already used as a replacement and has given better quality than the original approach. For example, the RDMD model [12] combines L2 loss with DMD and obtains results, often superior to one-way diffusion and OT-based models. In this work, there is a more high-quality loss: PatchNCELoss instead of L2, so much more attention is paid to preserving image features rather than just pixel-by-pixel proximity. This can improve the quality even further.

In practice, we have several problems to solve. Firstly, it is necessary to generalize the PatchNCELoss on UNet architectures used in the diffusions since the DMD method explicitly assumes the same generator and diffusion architectures. Secondly, old architectures used in CUT should be replaced with more modern and high-quality ones. Thirdly, we need to optimize the hyperparameters and make a valid comparison with baselines.

5 Experiments

This section evaluates the proposed method CUT+DMD on two unpaired image translation tasks. In Section 5.1, we assess performance on digit translation between classes 3 and 2 from the MNIST dataset, while Section 5.2 examines the translation between wild felines and domestic cats using the AFHQv2 dataset.

We use identical denoiser architecture and parameterization across all components: the target score model, fake score model, and generator. The pretrained target score is utilized in two ways: providing the starting weights for the fake score model, and initializing the generator

with fixed noise level of 1.0 to accommodate its single-step nature. The denoiser parameterization is trained to predict the clean images of the target domain, therefore, such initialization should significantly speed up convergence and nudge the model to use the information about the target domain more efficiently. During training, we optimize the generator once per every three updates of the fake score model.

To quantitatively assess the quality of the methods, we employ five standard metrics:

1 SSIM (Structural Similarity Index)

Measures structural similarity between generated and target images, focusing on luminance, contrast, and structure. Values range from 0 to 1, where higher values indicate better image structure preservation.

2 PSNR (Peak Signal-to-Noise Ratio)

Quantifies pixel-level fidelity by comparing the maximum possible signal power to noise (MSE). Expressed in decibels (dB), higher PSNR values denote lower pixel-wise distortion.

3 LPIPS (Learned Perceptual Image Patch Similarity) [16]

Evaluates perceptual similarity using feature extraction from pretrained networks. Lower LPIPS scores reflect better semantic alignment with the target domain.

4 FID (Fréchet Inception Distance) [4]

Compares the distributions of generated and real images in feature space (via Inception-v3). Lower FID values indicate closer alignment with the target distribution at a high level.

5 L2 (Euclidean distance)

Quantifies pixel-level similarity of the generated and target images.

5.1 MNIST

We compare the proposed method CUT+DMD with the baseline RDMD[12], which often obtains results superior to the one-way diffusion and OT-based models. Additionally, we modify the original CUT pipeline by removing its two-layer MLP applied to the image patches. We hypothesize that this component slows convergence, and our ablation study confirms that its removal improves training dynamics.

The results on the train MNIST data are represented in the table 5.1. Notably, our MLP-free variant achieves a comparable FID score, outperforming all baselines in both L2 and LPIPS

	SSIM	PSNR	LPIPS	L2	FID
RDMD	0.5513	14.4424	0.1755	10.8196	7.6910
CUT+DMD w/o MLP	**0.5920**	**15.0165**	**0.1598**	**10.2195**	**7.5804**
CUT+DMD w/ MLP	0.5555	14.4057	0.1740	10.9207	7.9357

Table 5.1: Metrics on the train MNIST data

metrics. This indicates that our generator produces images closer to the target distribution (class 2), both in terms of pixel-level and semantic similarity, without image quality degradation.

Theoretically, one might anticipate a slight increase in L2 error (reduced pixel-level accuracy) alongside a decrease in LPIPS (improved perceptual quality), as the model prioritizes high-level features over strict pixel matching. We hypothesize that this trade-off would become more pronounced on more complex datasets, where semantic consistency and stylistic preservation play a greater role.

	SSIM	PSNR	LPIPS	L2	FID
RDMD	0.5552	14.5601	0.1722	10.6958	**8.1010**
CUT+DMD w/o MLP	**0.5948**	**15.1277**	**0.1571**	**10.0972**	8.2204
CUT+DMD w/ MLP	0.5565	14.5240	0.1748	10.7784	8.1293

Table 5.2: Metrics on the test MNIST data

Table 5.2 shows consistent results, with the CUT+RDMD without MLP method maintaining its superior metrics at levels comparable to the training dataset results. This consistency between training and test performance indicates robust learning stability of the proposed architecture and also suggests that the model generalized effectively without overfitting to the training data.

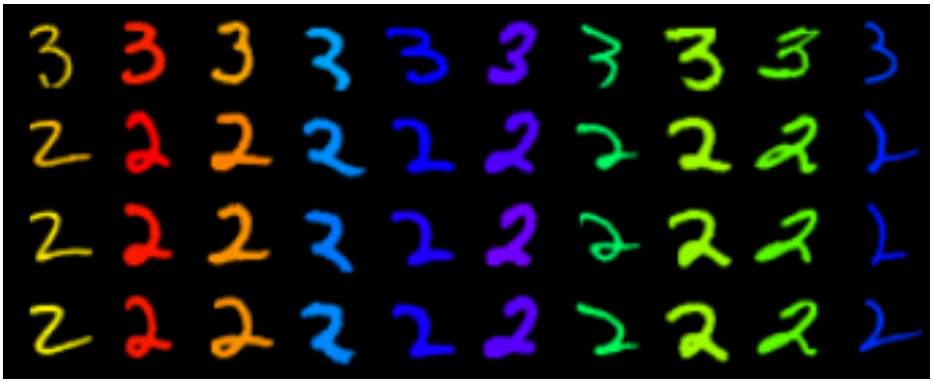

Figure 5.1: Generator outputs on MNIST train data: (1) Source, (2) RDMD, (3) MLP-free DMD+CUT, (4) DMD+CUT with MLP

Meanwhile, qualitative results can be observed in figure 5.1 for train MNIST data and figure 5.2 for test MNIST data. They demonstrate that while all evaluated methods achieve

plausible digit-to-digit translation, the MLP-free CUT+DMD model outperforms alternatives by striking an optimal balance between content accuracy and style fidelity. For instance, this method demonstrates superior preservation of digit stroke width (as in the seventh column of the figure 5.2) and angularity (as in the third column of the figure 5.2) compared to the alternative approaches, especially on the test data.

Figure 5.2: Generator outputs on MNIST test data: (1) Source, (2) RDMD, (3) MLP-free DMD+CUT, (4) DMD+CUT with MLP

5.2 Wild2Cat

As in the previous subchapter, we compare MLP-free CUT+DMD (as it gave better results on the MNIST data) with the baseline RDMD[12]. Given that RDMD combines DMD and L2 regularization term weighted by λ_{OT}, we explore different values of λ_{OT}.

	SSIM	PSNR	LPIPS	L2	FID
DMD+CUT w/o MLP, $\lambda_{\text{OT}} = 0.2$	0.445	15.75	0.394	18.62	24.12
DMD+CUT w/o MLP, $\lambda_{\text{OT}} = 1.0$	0.535	16.48	**0.357**	17.05	22.74
RDMD, $\lambda_{\text{OT}} = 0.05$	0.496	16.71	0.385	16.61	17.97
RDMD, $\lambda_{\text{OT}} = 0.1$	**0.544**	**17.35**	0.365	**15.43**	**17.82**

Table 5.3: Metrics on the test Wild2Cat data

The quantitative results for the test data are presented in table 5.3. Our analysis reveals that RDMD with $\lambda_{\text{OT}} = 0.1$ and MLP-free DMD+CUT with $\lambda_{\text{OT}} = 1.0$ achieve comparable performance in terms of perceptual similarity, indicating that both methods effectively preserve the style of input images. However, the DMD+CUT underperforms the RDMD on image quality metrics (PSNR and FID), suggesting its generated cat images appear less realistic.

We attribute this phenomenon to the inherent similarity between the domestic cat and wild feline domains. In such cases of domain proximity, increasing the regularization coefficient demonstrably improves both input-output similarity and FID scores. This consistent pattern is observable across all provided results tables.

Based on these findings, we hypothesize that increasing λ_{OT} could enable DMD+CUT to achieve FID scores comparable to RDMD.

Figure 5.3: Generator outputs on Wild2Cat train data: (1) Source, (2) MLP-free DMD+CUT, $\lambda_{\text{OT}} = 0.2$, (3) MLP-free DMD+CUT, $\lambda_{\text{OT}} = 1.0$ (4) RDMD, $\lambda_{\text{OT}} = 0.05$, (5) RDMD, $\lambda_{\text{OT}} = 0.1$

In the meanwhile, qualitative results are represented in the figures 5.3 for the train dataset and in 5.4 for the test dataset. While DMD+CUT achieves comparable LPIPS and SSIM scores and lower FID than RDMD, its outputs – particularly with $\lambda_{\text{OT}} = 1.0$ – demonstrate stronger structural alignment with input images. This manifests in two key aspects. Firstly, the method exhibits enhanced preservation of facial contour, as evidenced in the third column of the train figure 5.3 and the fourth column of the test figure 5.4. Secondly, it shows improved retention of facial expressions, visible in the fourth and tenth columns of the train 5.3 and the third and eighth columns of the test 5.4.

Figure 5.4: Generator outputs on Wild2Cat test data: (1) Source, (2) MLP-free DMD+CUT, $\lambda_{\text{OT}} = 0.2$, (3) MLP-free DMD+CUT, $\lambda_{\text{OT}} = 1.0$ (4) RDMD, $\lambda_{\text{OT}} = 0.05$, (5) RDMD, $\lambda_{\text{OT}} = 0.1$

6 Conclusion

For the thesis research, we developed an implementation of the proposed DMD+CUT method by modifying an existing codebase. This required substantial technical work including: rewriting training and evaluation scripts, implementing custom loss functions, and making novel architectural modifications to adapt the framework for our specific tasks. The implemented method was subsequently evaluated on several datasets. The results revealed two distinct outcomes: on the MNIST dataset, the approach demonstrated unambiguous improvements across all evaluation metrics, while on the Wild2Cat dataset it showed slightly inferior quantitative performance despite producing qualitatively superior visual results. Future research could explore adjusting the regularization coefficients, which might improve the FID metric, and validation on additional benchmark datasets.

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